



Formal language and automata

1

Introduction

Content

- What is Formal Language and automata?
- Mathematical preliminaries and notation
 - Sets
 - Function and relations
 - Graphs and trees
 - Proof techniques
- Three basic concept
 - Languages
 - Grammars
 - Automata

What is Formal Language and automata?

What is Formal Language and automata?

Formal languages define **what** strings belong

Automata define **how** machines recognize them.

Sets

- A **set** is a well-defined collection of distinct objects.

$$A = \{1, 2, 3, 4\}$$

- Objects inside a set are called **elements**.
- Memberships
 - x is an element of A
 - x is not an element of A

Sets

- Important Sets

- Natural Number

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}$$

- Integers

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

- Real Number

$$\mathbb{R}$$

- Subset : A set A is a subset of B if every element of A is also in B :

$$A \subseteq B$$

Sets

➤ Union

- $A \cup B$
- All elements in either A or B

➤ Intersection

- $A \cap B$
- Elements common to both sets

➤ Difference

- $A \setminus B$
- Elements in A but not in B

Sets

➤ Power Set

- The **power set** of A is the set of all subsets of A :
- $\mathcal{P}(A)$
- Example: $A = \{1, 2\}$

Functions and Relations

► Functions

- A function is a mapping:
- $f: X \rightarrow Y$
- Each element of X maps to exactly one element of Y .
- Example: $f(x) = x^2 + 2x + 1$

► Injective (One-to-One)

- Different inputs give different outputs.

► Surjective (Onto)

- Every element in the codomain has at least one preimage.

► Bijective

- Both injective and surjective. A perfect one-to-one correspondence.

Functions and Relations

- A relation is a subset of the Cartesian product:
 - $R \subseteq X \times Y$
 - Example: $R = \{(1, 2), (2, 3)\}$
- **Reflexive**
- $(x, x) \in R$
- **Symmetric**
- $(x, y) \in R \Rightarrow (y, x) \in R$
- **Transitive**
- $(x, y) \in R \wedge (y, z) \in R \Rightarrow (x, z) \in R$

Graph and trees

■ Graphs

- A graph is defined as: $G = (V, E)$

■ Types of Graphs

- Undirected graph:

$$(u, v) = (v, u)$$

- Directed graph:

$$(u, v) \neq (v, u)$$

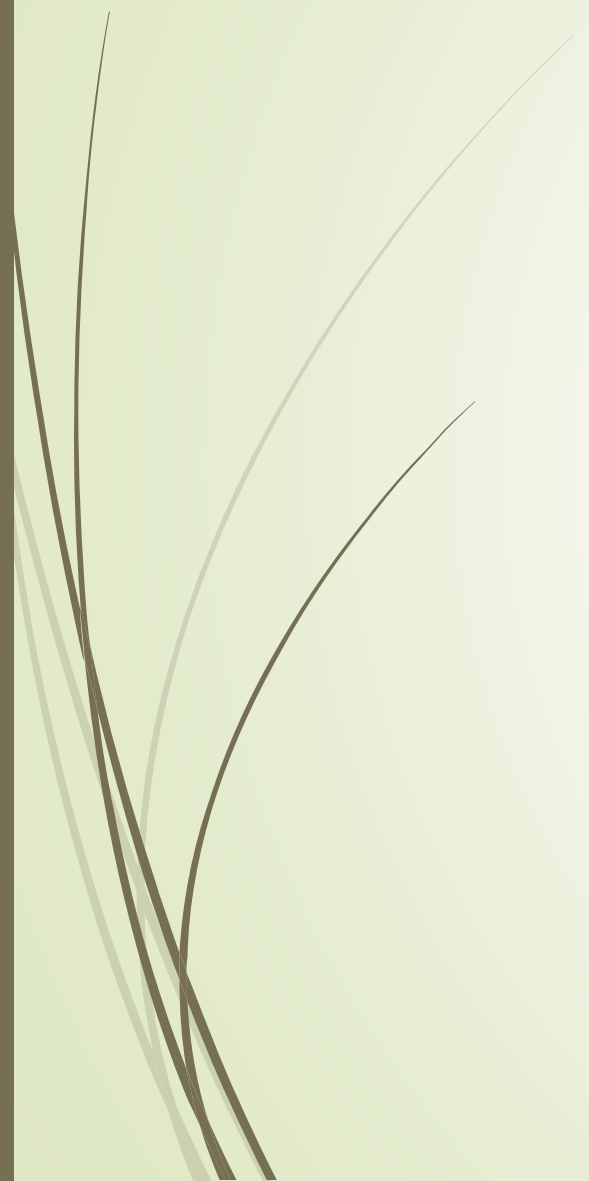
■ Trees

- A tree is a connected graph with no cycles.
- Properties:
- If a tree has n vertices, it has exactly $n - 1$ edges.
- There is a unique simple path between any two nodes.

Proof techniques

- Proof by induction
 - Base case
 - Inductive step
- Proof by contradiction
 - Assumption
 - Derivation
 - Conclusion

Think



Alphabet

- An **alphabet** is a finite, non-empty set of symbols.
- $\Sigma = \{a, b\}$
- Examples:
 - $\Sigma = \{0, 1\}$ (binary alphabet)
 - $\Sigma = \{a, b, c\}$

String

- ▶ A **string** is a finite sequence of symbols from an alphabet.
- ▶ Example:
 - ▶ If $\Sigma = \{0, 1\}$
 - ▶ $w = 0101$
- ▶ The **length** of string w is written as:
 - ▶ $|w|$
- ▶ Example:
 - ▶ $|0101| = 4$
- ▶ **Empty String** : The empty string is the unique string with length 0.
 - ▶ ϵ
 - ▶ $|\epsilon| = 0$

Kleene Star

- The Kleene star of an alphabet Σ is the set of all possible finite strings over Σ .
- Σ^*
- Example:
 - If $\Sigma = \{a, b\}$
 - $\Sigma^* = \{\epsilon, a, b, a, a, a, b, b, abbaaa \dots\}$

Formal language

- A **formal language** is any subset of Σ^* .
 - $L \subseteq \Sigma^*$
- Example:
 - Let $\Sigma = \{0, 1\}$
 - $L = \{w \in \Sigma^* \mid w \text{ ends with } 1\}$
- So:
 - $L = \{1, 01, 101, 0001, \dots\}$

Automata

- An automata is an abstract model of a digital computer
- **Finite Automaton Definition**
- A **finite automaton** is formally defined as a 5-tuple:
- $M = (Q, \Sigma, \delta, q_0, F)$
- Where:
 - Q : finite set of states
 - Σ : input alphabet
 - δ : transition function
 - $q_0 \in Q$:start state
 - $F \subseteq Q$: set of accepting states

Thank You

Quiz1

- $A=\{1,2,3,4\}$ $B=\{3,4,5,6\}$
 - Carilah Union, intersection, difference, dan powerset
- $R=\{(1,5),(2,4),(2,6),(3,5)\}$
 - Carilah domain dan range
- $f(x)=2x+3$, dengan domain $X=\{1,2,3,4\}$
 - Berapa hasilnya?